

eQMS - UI Design

1. General Principles

1. Multi-Tenant Awareness

- ~~Tenant selector or branding if users can belong to multiple tenants.~~
- User sees only their tenant's documents and users.

2. Role-Based Views

- UI adapts to the user role (Viewer, Editor, Approver, Admin, Auditor).
- Field-level ABAC rules hide sensitive fields dynamically.

3. Audit Visibility

- Users see only audit events they are authorized for.
- Audit trail access is **restricted to Auditor/Admin roles**.

4. Compliance First

- Actions should be logged automatically.
- Provide **warnings for sensitive actions** (e.g., deleting a document).

5. Responsive & Scalable

- Grid/list views for documents, users, audit events.
- Filters and search optimized for large datasets.

2. Main UI Modules

2.1 Dashboard

- Summary of:
 - Recent documents created/updated
 - Pending approvals
 - Recent audit events (limited, for auditors/admins)
 - Quick access to tenants or users

2.2 Document Management

- **List/Grid View**
 - Columns: Document ID, Title, Version, Owner, Status, Classification, Last Updated
 - **Filters:** Classification, Owner, Status, Date range
 - ABAC filters applied to hide unauthorized documents
- **Actions**
 - View metadata

- View/download content (if permitted)
- Edit/Update (Editor/Approver)
- Approve (Approver)
- Delete/Archive (Admin)

- **Document Detail Page**

- Metadata
- Versions (historical)
- Field-level visibility (ABAC applied)
- Audit trail link (if permitted)
- Workflow/approval status

2.3 Users & Roles

- List of users with tenant, roles, last login
- Actions: Create/Edit/Delete user (Admin only)
- Role assignment UI:
 - Dropdown or checkbox list of roles
 - Shows inherited permissions
- View role definitions and associated permissions

2.4 Tenants

- Multi-tenant admin view:
 - List of tenants with number of users/documents
 - Create/Edit/Delete tenants (Admin only)
- Assign users to tenants

2.5 Audit Trail

- Grid/List view for audit events
 - Columns: Timestamp, User, Role, Action, Object Type, Object ID, Fields Returned, Status
- Filters:
 - By user, tenant, action, date range, object type
 - Field-level filters optional
- Export feature: CSV/JSON
- Detail view:
 - Shows full payload (before/after) for CUD operations
 - Fields returned for read actions

- Request URL and correlation ID for traceability

2.6 Notifications / Alerts

- Optional alerts for:
 - Unauthorized access attempts
 - Failed audit logging
 - Document approvals pending

3. Interaction Patterns

1. Grid + Filters

- All lists (documents, users, audit) support column filtering, sorting, and pagination.

2. Modal Dialogs

- For creating/editing documents, users, tenants
- Approvals can be modal to avoid page navigation

3. Expandable Rows

- For document list: expand to show versions and field-level visibility

4. Audit Drill-down

- Click on an audit event to see full details, including fields returned

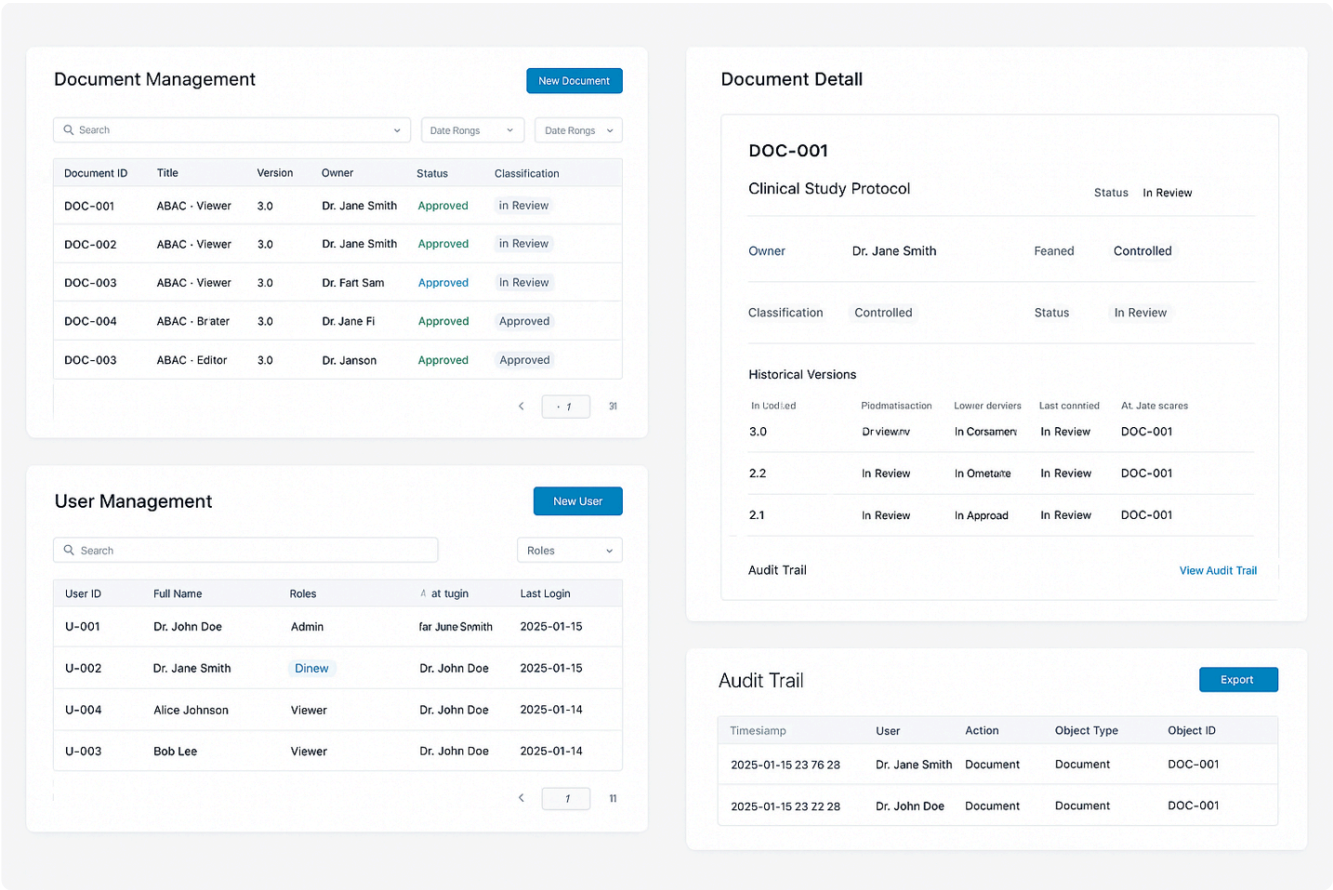
5. Dynamic Field Masking

- UI automatically hides fields based on ABAC evaluation
- Prevents accidentally showing restricted data

4. Security & Compliance Features in UI

- **Role-based menus** – users only see actions permitted by RBAC/ABAC
- **Immutable actions log** – every action triggers an audit event
- **Field-level masking** – sensitive fields hidden, consistent with backend ABAC
- **Download/export controls** – restricted by permission
- **Session timeout / inactivity lock** – ensures secure access
- **Request confirmation for destructive actions** – delete/archive

5. Concept



5. Concept

Simple tables, for now, without Search capabilities, only filtering.

Document Management

New Document

Q Search	Date Rangs	Date Rangs			
Document ID	Title	Version	Owner	Status	Classification
DOC-001	ABAC - Viewer	3.0	Dr. Jane Smith	Approved	In Review
DOC-002	ABAC - Viewer	3.0	Dr. Jane Smith	Approved	In Review
DOC-003	ABAC - Viewer	3.0	Dr. Fait Sam	Approved	In Review
DOC-004	ABAC - Brater	3.0	Dr. Jane Fi	Approved	Approved
DOC-003	ABAC - Editor	3.0	Dr. Janson	Approved	Approved
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User Management

New User

Q Search				Roles	
User ID	Full Name	Roles	At tugin	Last Login	
U-001	Dr. John Doe	Admin	far June Smith	2025-01-15	
U-002	Dr. Jane Smith	Dinew	Dr. John Doe	2025-01-15	
U-004	Alice Johnson	Viewer	Dr. John Doe	2025-01-14	
U-003	Bob Lee	Viewer	Dr. John Doe	2025-01-14	
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Document Detail

DOC-001

Clinical Study Protocol

Status In Review

Owner Dr. Jane Smith Feaned Controlled

Classification Controlled Status In Review

Historical Versions

In Lodied	Piodmatisation	Lower derviers	Last contried	At. Jate scates
3.0	Dr viewrv	In Corsamen	In Review	DOC-001
2.2	In Review	In Ometate	In Review	DOC-001
2.1	In Review	In Approad	In Review	DOC-001

Audit Trail View Audit Trail

Audit Trail

Export

Timesiamp	User	Action	Object Type	Object ID
2025-01-15 23 76 28	Dr. Jane Smith	Document	Document	DOC-001
2025-01-15 23 72 28	Dr. John Doe	Document	Document	DOC-001